Katarzyna Wręczycka, Ph.D.

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Profile

Computational Biologist (PhD) specializing in machine learning and deep learning for cancer biology and translational disease research, with expertise in spatial transcriptomics, single-cell RNA-seq, integrative multi-omics analysis. Thrives in creative and dynamic international environments, with high collaboration between teams.

Experience

02.2023 – 04.2026

Bioinformatician & Machine Learning Scientist

remote from Berlin, Germany

Engaged in B2B contracts with pharmaceutical companies, start-ups and R&D consultancy companies. Key highlights include:

- **Oncology biomarker discovery in healthcare**:
  - **Machine learning for drug development:** Applied positive unlabeled learning with random forests and SVM (with pulearn) and autoencoder (in PyTorch) to identify transcription factors as drug targets, integrating clinical trial insights (Merck, Oncology Data Science Department & Ardigen).
  - **Translational oncology signature discovery:** Identified cGAS-STING pathway, IFN-γ, NK cell, and interleukin biomarkers driving treatment response in cancer patients with limited therapeutic options through close collaboration with clinical and experimental teams. Performed survival analysis to guide treatment stratification using gene expression, gene mutation and disease-drug associations data from TCGA, TEMPUS, GenomOncology and internal clinical trial databases (Merck, Oncology Bioinformatics Department & Ardigen).
  - **Single-cell immuno-oncology profiling:** Characterized the cGAS-STING pathway and tumor microenvironment response to selected drugs in cancer based on scRNA-seq mouse models (R/Seurat)(Merck, Oncology Bioinformatics Department & Ardigen).
- **Omics data analysis of neurodegenerative reprogramming:** Investigated glia-to-neuron reprogramming in Alzheimer's disease through multi-omics analysis (RNA-seq, ATAC-seq, H3K4me2 and H3K27me3 ChIP-seq, BS-seq) using R/Bioconductor. Performed motif enrichment, pathway analysis, and applied deep learning-based local DNA sequence analysis using the Enformer model (Avsec et al., Nature Methods, 2021) to identify and characterize target genes and regulatory enhancers (Stardustries & Ardigen).
- **Menstrual blood biomarker discovery**: Led a pilot project on classifying endometriosis, blood profiling, oncology biomarker discovery and immune cell dynamics using targeted RNA-seq and scRNA-seq from menstrual effluent (scanpy, scvi-tools) and XGBoost for feature importance analysis performed on AWS (EC2, S3) (Start-up theblood).
- **Infrastructure and workflow automation:** Built Docker (compose) images, deployed workflows in Nextflow on AWS and Ardigen's Kubernetes cluster.
- **Detection of PFAS for water safety monitoring**: Built deep learning model (VAE, PyTorch) to detect spikes of "forever" chemicals from a patented device (link); contributed to Kafka-based streaming pipeline (Aqua-Q).
- **Software development for web applications:** Enhanced IGV app browser for genomic data visualization and integrated in-house and publicly-available sequencing (epi)genomic data (incl. ENCODE) using JavaScript/JQuery; in collaboration with UX/UI designer (Novo Nordisk & Ardigen).

Selected software & method contributions (public, independent):

- **Diffusion models for immune cell images:** Built a generative UNet-based diffusion model with linear noise scheduling for H&E-stained immune cell image inpainting, with an interactive Streamlit frontend for model exploration (GitHub link).

- **Immune cell classifier**: Developed and trained a CNN-based image classifier for immune cell detection in H&E-stained histopathological images using Xception with an MLP head; optimized and exported to TensorFlow Lite, containerized with Docker, and deployed on AWS Lambda for scalable inference (GitHub link).
- **Contribution to nf-core/spatialxe Xenium workflow:** Contributed to the community-maintained nf-core spatial transcriptomics 10X Xenium workflow (PR #130) that adds SCS segmentation method designed for Stereo-seq and Seq-scope.
- **Light-weight spatial transcriptomics platform:** An interactive Django-based spatial transcriptomics platform (Visium) that visualizes gene expression with an LLM-powered chatbox (Ollama), deployed via Streamlit frontend, containerized with Docker (GitHub link).

04.2025 – 06.2025

Bioinformatician | Research Assistant II

ETH Zürich & Functional Genomics Center Zürich (FGCZ) of the University of Zurich (UZH)
Employed by the Prof. von Meyenn Lab (40% position), Prof. Bar-Nur Lab (40%) and FGCZ/UZH (20%)

- **Gene regulation analysis and data visualization:** across muscle aging with natural compounds (RNA-seq), germline chimerism (snRNA-seq), yo-yo effect in adipocytes (RNA-seq, CUT&TAG) using R/Bioconductor, and assembly of Y chromosome of *Mus minutoides* (WGS).
- **Nextflow Tower for NGS:**
  - Maintained and operated end-to-end Nextflow pipelines (RNA-seq, CUT&TAG); deployed via Nextflow Tower on SLURM (ETH Zürich cluster), with containerization using Docker/Singularity and environment management via Spack.
- **Interactive web-based genomic visualizations:**
  - Deployed interactive R Shiny apps via ShinyProxy for single cell RNA-seq data available at FGCZ servers shiny-public.fgcz.uzh.ch (snRNA-seq: MHUO, Mouse epiAT memory, Human AT memory)
- **Cross-disciplinary collaboration** with wet-lab scientists to interpret analysis and experiments.

10.2015 – 11.2021

PhD Student in Computational Molecular Biology

Berlin Institute for Medical Systems Biology (BIMSB) at the Max Delbrück Center for Molecular Medicine (MDC), Berlin, Germany.

- **Genomic artifact detection**: Investigated high-density transcription factor binding sites and identified spurious ChIP-seq peaks linked to RNA structures called DNA-RNA hybrids (R-loops), and hypo-methylated regions using elastic net regression and PCA [2]. Integrated data from publicly available resources, including ENCODE and Roadmap Epigenomics, as well as experimental datasets from ChIP-seq, ATAC-seq, Bisulfite-seq, and DRIP-seq/RDIP-seq studies.
- **DNA methylation biomarkers for acute coronary syndrome discovery**: Studied Bisulfite-seq data from patient-derived heart tissue and plasma, uncovering gene and transcription factor associations. My contribution included preprocessing raw Bisulfite-seq data, calling differentially methylated regions, associating them with genes, and transcription factors in cardiovascular diseases [1].
- Co-authored a peer-reviewed **review on statistical methods for analyzing Bisulfite-seq data** (accompanied by publicly available code at github.com/BIMSBbioinfo/Strategies_for_analyzing_BS-seq) [4].
- **Snakemake pipeline for Bisulfite-seq**: A scalable pipeline using TrimGalore!, BWA-meth/Bismark, samblaster, MethylKit/methylDackel, and genomation R packages, published as part of the PiGX: Pipelines in Genomics project [3].
- **Multi-omics analysis of high-risk pediatric neuroblastoma**: Identified DNA methylation biomarkers and transcription factor networks specific to high-risk cases from solid tumors and cell-free DNA in urine. Applied machine learning methods (incl. elastic net, ridge regression, XGBoost) and integrated patient-derived samples from Bisulfite-seq, RNA-seq, and ChIP-seq datasets in collaboration with Dr. med. Johannes H. Schulte Charité Hospital in Berlin.

03 – 09.2015

Visiting Predoctoral Researcher

Berlin Institute for Medical Systems Biology (BIMSB) at the Max Delbrück Center for Molecular Medicine (MDC), Berlin, Germany.
Awarded a €7,000 scholarship by the Institute of Computer Science, Polish Academy of Sciences (IPI PAN), co-financed by the European Union. Supervised by Dr. Altuna Akalin.

- **Maintained and co-developed the Bioconductor R package genomation**, designed for the annotation, visualization, and summary of genomic data (github.com/BIMSBbioinfo/genomation).

- Investigated phantom peaks (false positives) in publicly available ChIP-seq datasets, contributing foundational work later expanded during Ph.D. research [2].

Education

10.2015 – 05.2021

Ph.D. in Computational Molecular Biology

Humboldt University, Department of Biology, Berlin, Germany & Berlin Institute for Medical Systems Biology at the Max Delbrück Center for Molecular Medicine, Berlin, Germany

Dissertation: "Integrative analysis on the effect of DNA methylation on gene regulation in tumor and healthy tissues" (PDF). Advisors: Dr. Altuna Akalin and Prof. Dr. Uwe Ohler

10.2012 – 06.2014

M.Sc. in Bioinformatics and Systems Biology

University of Warsaw, Department of Mathematics, Informatics and Mechanics, Warsaw, Poland

Dissertation: "Genome-wide analysis of noncoding copy-number variations in human genome". Advisor: Prof. Dr hab. Anna Gambin

10.2010 – 06.2012

B.Sc. in Bioinformatics and Systems Biology

University of Warsaw, Department of Mathematics, Informatics and Mechanics, Warsaw, Poland

Dissertation: "Potential regulatory region discovery method implementation". Advisor: Dr hab. Bartosz Wilczynski. I developed an algorithm based on Bernoulli distribution to identify conserved noncoding regulatory regions (github.com/katwre/bioinf/bio_motif_ensembl)

Technical skills

- **Sequencing & Imaging Data**: scRNA-seq/snRNA-seq, spatial transcriptomics (10X Visium, Xenium), RNA-seq, ChIP-seq, CUT&Tag, Bisulfite-seq (WGBS/RRBS), WGS.
- **Programming & Tools:** In-depth knowledge of statistical programming, data analysis, and visualisations in R (CRAN/Bioconductor), Python (Biopython, NumPy, pandas, scikit-learn, seaborn, PyTorch); experienced in bash, and Javascript; unit tests (R/testthat; Python/unittest/pytest)
- **Machine Learning & Statistical Analysis:** Statistical analysis of (epi)genetic and imaging data, including:
  - statistical tests (e.g., t-tests, Wilcoxon tests), regression models (e.g., multivariate logistic regression, elastic net, Cox proportional hazards model), and classification algorithms (e.g., random forest, SVM, kNN); unsupervised learning (e.g., hierarchical clustering, K-means, EM algorithm) and dimensionality reduction (e.g., PCA, t-SNE);
  - applied deep learning approaches (variational autoencoders, CNNs, Transformer-encoder),
  - NLP (RAG, LLMs (LangChain, Mistral, LLaMA), vector search (Qdrant)).
  - privacy-preserving ML - federated learning (Flower)
- **High-Performance Computing:** HPS(Grid Engine and SLURM), Kubernetes, AWS (EC2, S3, Lambda, Beanstalk), Database Engines (MySQL, SQLite); Docker (compose)/Singularity containerization
- **Version Control and Software Management:** GNU/Linux operating system, git version control system, and conda and GNU Guix package management systems
- **Web development basics:** Django, FastAPI, Javascript/JQuery, HTML/CSS

Soft Skills:

- Interdisciplinary collaboration, problem solving, ability to explain complex ideas to diverse audiences

References

Dr. Altuna Akalin
Head of Bioinformatics & Omics Data Science Platform
The Max Delbrück Center for Molecular Medicine (MDC), the Berlin Institute for Medical Systems Biology (BIMSB)
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Dr. Jan Majta (Line manager at Ardigen)
Director of AI Solutions
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Dr. Olga Bogarytova
Data scientist, Translational Biomarker Lead, RU Oncology
Merck KGaA
Frankfurter Str. 250, 64293 Darmstadt, Germany
olga.bogatyrova@merckgroup.com

Prof. Dr. Ferdinand von Meyenn (40% employment at ETH Zurich)
Associate Professor
Department of Health Sciences and Technology at ETH Zurich
ferdinand.vonmeyenn@hest.ethz.ch

Dr. Bora Uyar
Staff Bioinformatics Scientist
The Max Delbrück Center for Molecular Medicine (MDC), the Berlin Institute for Medical Systems Biology (BIMSB)
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Publications

[1] R. R. C. Cuadrat, A. Kratzer, H. G. Arnal, R. A. C., K. Wreczycka, A. Blume, V. Ebenal, T. Mauno, B. Osberg, M. Moobed, J. Hartung, C. Seppelt, D. Meteva, A. Haghikia, D. Leistner, U. Landmesser and A. Akalin, "Cardiovascular disease biomarkers derived from circulating cell-free dna methylation," NAR Genomics and Bioinformatics, 2023. DOI: 10.1093/nargab/lqad061.

[2] K. Wreczycka, V. Franke, B. Uyar, R. Wurmus, S. Bulut, B. Tursun and A. Akalin, "HOT or not: examining the basis of high-occupancy target regions," Nucleic Acids Research, vol. 47, no. 11, pp. 5735–5745, May 2019. DOI: 10.1093/nar/gkz460.

[3] R. Wurmus, B. Uyar, B. Osberg, V. Franke, A. Gosdschan, K. Wreczycka, J. Ronen and A. Akalin, "PiGx: reproducible genomics analysis pipelines with GNU Guix," GigaScience, vol. 7, no. 12, Oct. 2018, giy123. DOI: 10.1093/gigascience/giy123.

[4] K. Wreczycka, A. Gosdschan, D. Yusuf, B. Grüning, Y. Assenov and A. Akalin, "Strategies for analyzing bisulfite sequencing data," Journal of Biotechnology, vol. 261, pp. 105–115, 2017, Bioinformatics Solutions for Big Data Analysis in Life Sciences presented by the German Network for Bioinformatics Infrastructure. DOI: 10.1016/j.jbiotec.2017.08.007.

Conferences & workshops

- attendance, AI Dev Tools Zoomcamp by DataTalks.Club, Nov 2025 – Jan 2026, online (GitHub link)
- attendance, Machine Learning Zoomcamp by DataTalks.Club, Oct 2025 – Jan 2026, online (GitHub link)
- attendance, Data Engineering Zoomcamp by DataTalks.Club, Jan 2026 – April 2026, online (GitHub link)
- attendance, Introduction to GPU Programming for Life Sciences, Swiss Institute of Bioinformatics workshop, 11–12 Feb 2026, Switzerland
- attendance, Ensuring More Accurate, Generalisable, and Interpretable Machine Learning Models for Bioinformatics, Swiss Institute of Bioinformatics workshop, 17 Nov. 2025, Switzerland (GitHub link)
- attendance, Using Large Language Models for Biodata Exploration: From Theory to Practice, Swiss Institute of Bioinformatics workshop, 30 Sept. 2025, Zurich, Switzerland (GitHub link)
- attendance, PyData Berlin Conference and Workshops, 1–3 Sep. 2025, Berlin, Germany
- attendance, Federated Learning in Bioinformatics, Swiss Institute of Bioinformatics workshop, 29 Aug. 2025, Lugano, Switzerland (GitHub link)
- speaker, "Assessing prognostic and diagnostic features of neuroblastoma methylome", DKTK - Young Academics Meeting, 6 Sep. 2019, Berlin Germany
- attendance, EMBL Conference: Cancer Genomics, 4-6 Nov. 2019, Heidelberg, Germany
- speaker, "Assesing prognostic and diagnostic features of high-dimentational (epi)genomics data for neuroblastoma", RECOMB, 5-8 May 2018, Washington, D.C., USA
- Europython conference and workshops, 10-16 July 2017, Rimini, Italy
- speaker, "HOT or not: Examining the basis of high-occupancy target regions", CRG-BIMSB MDC PhD Retreat, 15-18 Mai 2017, Girona, Spain
- EMBL-EBI Cancer Genomics Course, 20-23 June 2017, Hinxton, United Kingdom
- poster presentation "HOT or not: redefining the origin of high-occupancy target regions", EMBL conference From Functional Genomics to Systems Biology, 12-15 Nov. 2016, Heidelberg, Germany
- speaker, "Genomation: an R package to summarize, annotate and visualize genomic intervals", Bioinformatics Social Meetings Berlin, 16 Sep. 2016, Berlin Germany
- poster presentation "Genomation: an R package to summarize, annotate and visualize genomic intervals", the Max-Planck-Gesellschaft Otto Warburg Summer School, 6-11 Sep. 2015, Berlin, Germany
- Europython conference and workshops, 21-27 July 2014, Berlin, Germany

Dr. Katarzyna Wreczycka
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April 30th, 2026

Dear Prof. Bentires-Alj,

I am excited to apply for the Postdoc position in your Tumor Heterogeneity, Metastasis and Resistance lab at the Department of Biomedicine in Basel. Holding a Ph.D. in Computational Molecular Biology from the Bioinformatics and Omics Data Science Platform at the Berlin Institute for Medical Systems Biology (MDC-BIMSB), combined with industry experience, I have a robust background in machine learning and deep learning methods for next-generation sequencing, scRNA-seq, spatial transcriptomics. I am eager to contribute to the wide range of basic and translational research projects in your teams, leveraging my technical skills and collaborative approach to advance research in molecular and cellular mechanisms of breast cancer and metastasis.

During my doctoral research, I collaborated with interdisciplinary teams, including medical doctors at Charité Hospital, wet-lab scientists at the MDC-BIMSB, and bioinformaticians at the Berlin Institute of Health, to identify DNA methylation biomarkers based on Bisulfite-seq for pediatric cancer neuroblastoma and cardiovascular diseases from blood samples. This work combined machine learning and statistical methods with reproducible tools and workflows for next generation sequencing (NGS) data analysis developed in R/BioC and Python. In 2023, I transitioned to industry as a freelance bioinformatics and data science consultant, primarily through Ardigen, where I collaborated with pharmaceutical companies and startups such as Merck's Oncology Data Science Department and Novo Nordisk. These projects strengthened my ability to work across disciplines, collaborating with bioinformaticians, data scientists, wet-lab researchers and software engineers. Highlights of my recent work include identifying biomarkers driving treatment response in the cGAS-STING pathway and related immune mechanisms (IFN-γ, NK cell, and interleukin signaling) in cancer patients from scRNA-seq and NGS data, and nominating novel therapeutic targets and potential drug candidates for clinical trials using machine learning methods.

While consulting has been a rewarding way to contribute to diverse projects, I am eager to further develop as an independent scientist and engage in basic and translational research projects. I am particularly drawn to your laboratory's innovative focus on understanding tumor heterogeneity and metastatic resistance, and the opportunity to apply single-cell, spatial transcriptomics, and other modalities combined with machine learning to uncover the molecular mechanisms driving treatment escape.

I would welcome the opportunity to discuss how my expertise in single-cell, spatial transcriptomics and other modalities combined with machine learning approaches, could contribute to uncovering the mechanisms of tumor heterogeneity and metastatic resistance in your laboratory. I look forward to hearing from you.

Sincerely,

Dr. Katarzyna Wreczycka

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Dr. Altuna Akalin

Bioinformatics and Omics Data Science

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Berlin, 31.10.2025

Reference letter for Dr. Katarzyna Wręczycka

To whom it may concern,

I am pleased to recommend Dr. Katarzyna Wręczycka, who completed her PhD under my supervision at the Berlin Institute for Medical Systems Biology, Max Delbrück Center for Molecular Medicine. During her doctoral research, Katarzyna demonstrated outstanding analytical and computational skills, particularly in applying and developing machine-learning approaches to address complex questions in genomics and epigenomics.

Katarzyna's PhD research focused on machine learning applications in computational molecular biology, where she carried out large-scale regulatory modeling on heterogeneous datasets using approaches such as regression, classification, dimensionality reduction, and model interpretation. Her work aimed to uncover biologically meaningful structure in high-dimensional data, combining rigorous computational modeling with biological insight. In particular, she analyzed high-occupancy target (HOT) regions in ChIP-seq data and performed genome-wide DNA-methylation profiling in neuroblastoma tumors and Bisulfite-seq liquid-biopsy samples from patients with acute coronary syndrome, in collaboration with clinical teams at the Charité Hospital and the Berlin Institute of Health.

In addition to her analytical work, Katarzyna made significant contributions to method and software development. She co-developed genomation, an R/Bioconductor package for integrative genomic analysis, and designed a Snakemake pipeline for Bisulfite-seq data processing, implementing best practices for reproducibility, version control, and modularity. These efforts reflect her commitment to collaborative, reproducible, and scalable research, her strong software-engineering mindset, and her ability to produce tools of lasting value for the research community.

Katarzyna approaches scientific questions with creativity and analytical depth, combining conceptual thinking with careful implementation. She works with a high degree of independence, yet remains highly engaged and collaborative within a team. Her curiosity, clarity in communication, and reliability made her an integral part of the group, and she interacted thoughtfully and productively with colleagues from diverse scientific backgrounds.

I am confident that Katarzyna will drive advances in machine-learning approaches for complex biological and medical data. She brings both technical depth and scientific perspective, which make her well-suited for interdisciplinary research at the interface of computation and the life sciences. If you have any questions, please do not hesitate to contact me.

Sincerely,

Dr. Altuna Akalin
Group leader
Bioinformatics & Omics Data Science Platform
Berlin Institute for Medical Systems Biology at the Max-Delbrück-Center for Molecular Medicine

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Zurich, 2 September 2025

RE: Reference letter for Katarzyna Wreczycka

To whom it may concern

I am pleased to write this reference letter for Katarzyna Wreczycka, who worked in my group for Nutrition and Metabolic Epigenetics at ETH Zurich (epigenetics.ethz.ch) from April to June 2025, with a 40% appointment.
During her tenure, Kasia contributed to some of our current projects. She helped in the analysis of both RNA-seq and CUT&RUN datasets derived from adipose tissue, supporting our investigation into the epigenetic memory of obesity after weight loss. Kasia was also tasked to maintain and update the group's bioinformatics workflows on the ETH Euler cluster, including Nextflow Tower, SLURM, Singularity, and Spack.
In addition to her analytical work, Kasia provided technical support for our computational infrastructure and managed a successful migration of a Shiny application to ShinyProxy, hosted on the Functional Genomics Center Zürich servers (https://shiny-public.fgcz.uzh.ch/).
Her ability to quickly learn new technologies, work autonomously, was valued by her colleagues. Kasia consistently demonstrated professionalism, structurer working, and dedication, attributes which I am confident will allow her to thrive in any future role.
I strongly recommend Katarzyna Wreczycka for any opportunity that values computational and bioinformatics skills, and independent problem-solving. Please do not hesitate to contact me for further information.
Yours sincerely,

Prof. Dr. Ferdinand von Meyenn

HUMBOLDT-UNIVERSITÄT ZU BERLIN

URKUNDE

Die Lebenswissenschaftliche Fakultät verleiht

FRAU KATARZYNA WRĘCZYCKA

geboren am 19.07.1990 in Warschau, Polen

den akademischen Grad

DOCTOR OF PHILOSOPHY (Ph.D.)

nachdem sie ihre wissenschaftliche Befähigung im Fach

Biologie / Molekularbiologie

nachgewiesen hat.

Thema der Dissertation:

Integrative analysis on the effect of DNA methylation on gene regulation in tumor and healthy tissues

Die mündliche Prüfung fand am 13.09.2021 statt.

Für die Gesamtleistung wurde das Prädikat **magna cum laude** erteilt.

Berlin, den 13.09.2021

Prof. Dr.-Ing. Dr. Sabine Kunst

Prof. Dr. Dr. Christian Ulrichs